E2 Problems: complex numbers

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Give exact answers unless asked otherwise. Arguments in radians.

Practice

1. Write in the form $a+bi$:

(a) $(3+2i)(1-4i)$ (b) $\dfrac{2-i}{1+3i}$ (c) i^{2031}

2. Solve $z^2-4z+13=0$.

3. Find the modulus and the principal argument of

(a) $-1+i\sqrt{3}$ (b) $-3-3i$

4. Write $4e^{i\pi/6}$ and $2e^{-3i\pi/4}$ in the form $a+bi$.

5. Sketch in the complex plane:

(a) $|z-2i|=3$
(b) $|z-1|=|z+i|$
(c) $\arg z=\frac{\pi}{4}$
(d) $|z-1|\le|z+1|$

6. Compute $(1+i)^{10}$ using polar form.

7. Solve $z^3=8i$. Give the solutions in polar form and in the form $a+bi$, and draw them.

8. Prove for all $z,w\in\mathbb{C}$: (a) $z\bar{z}=|z|^2$; (b) $\overline{zw}=\bar{z}\,\bar{w}$; (c) z is real iff $z=\bar{z}$.

9. Find all real a,b with $(a+bi)^2=5-12i$.

10. The point $4+i$ is rotated by $90°$ anticlockwise about $1+i$. Find the image.

Exam-level

11. (a) Use De Moivre's theorem to express $\cos 3\varphi$ and $\sin 3\varphi$ in terms of $\cos\varphi$ and $\sin\varphi$.

(b) Solve $4x^3-3x=\frac{1}{2}$ exactly, by substituting $x=\cos\varphi$.

12. Given $P(z)=z^3-3z^2+7z-5$.

(a) Show that $z=1$ is a root.
(b) Find all roots of P.
(c) Explain without calculation why the non-real roots are each other's conjugates.

13. (a) Solve $z^4=-4$.

(b) Hence factor z^4+4 into two quadratics with real coefficients.

14. Let $\omega=e^{2\pi i/5}$.

(a) Show that $1+\omega+\omega^2+\omega^3+\omega^4=0$.
(b) Deduce that $\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}=-\frac{1}{2}$.
(c) (harder) Put $c=\cos\frac{2\pi}{5}$. Use $\cos\frac{4\pi}{5}=2c^2-1$ to find c exactly.

15. Solve $x^3+6x-20=0$ with Cardano's method. Find all three (complex) solutions.

16. (Bombelli) Cardano's formula for $x^3=15x+4$ gives $x=\sqrt[3]{2+11i}+\sqrt[3]{2-11i}$.

(a) Show that $(2+i)^3 = 2+11i$.
(b) Explain how this gives the solution $x = 4$, and find the other two real solutions.

17. Describe and sketch the set of all z with $|z-3| = 2|z|$.

18. Points 0, z and w (all distinct) form an equilateral triangle with vertices in anticlockwise order.

(a) Explain why $w = e^{i\pi/3}z$.
(b) Show that $z^2 + w^2 = zw$.

19. Given $f(z) = (1+i)z + 2$.

(a) Find the fixed point of f.
(b) Describe f geometrically.
(c) Find $f(f(f(f(0))))$ by thinking geometrically, then check by computation.

20. Use $e^x \cos x = \text{Re}\left(e^{(1+i)x}\right)$ to find $\displaystyle\int e^x \cos x \, dx$. Check by differentiating.

Challenge

21. Let $n \geq 2$ and $\omega = e^{2\pi i/n}$.

(a) Explain why $z^{n-1} + z^{n-2} + \cdots + z + 1 = (z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1})$.
(b) Deduce $\displaystyle\prod_{k=1}^{n-1}(1-\omega^k) = n$.
(c) Show $|1-e^{i\theta}| = 2\sin\frac{\theta}{2}$ for $0 < \theta < 2\pi$, and deduce $\displaystyle\prod_{k=1}^{n-1} 2\sin\frac{k\pi}{n} = n$. (Geometric meaning: the product of the lengths of the diagonals and two sides from one vertex of a regular n-gon inscribed in the unit circle is n.)

22. Let $\omega = e^{2\pi i/3}$. Prove that the triangle with vertices a, b, c is equilateral if and only if

$$a^2 + b^2 + c^2 = ab + bc + ca.$$

(Hint: show $(a+\omega b+\omega^2 c)(a+\omega^2 b+\omega c) = a^2+b^2+c^2-ab-bc-ca$, and interpret $a+\omega b+\omega^2 c = 0$.)

23. Prove for θ not a multiple of 2π:

$$\cos\theta + \cos 2\theta + \cdots + \cos n\theta = \frac{\sin\frac{n\theta}{2}\,\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}.$$

(Hint: it is the real part of a geometric sum, and use $e^{ix} - 1 = e^{ix/2}\cdot 2i\sin\frac{x}{2}$.)

24. Prove that all solutions of $(z+1)^5 = z^5$ lie on the line $\text{Re}\, z = -\frac{1}{2}$. Then find them exactly.

25. (a) Explain why "i^i" can be given the real value $e^{-\pi/2}$.
(b) Why does i^i have infinitely many possible values? Are they all real?